Akoizo Original Practice - Form 1

Division B | Scrambler | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which wins: a lower or higher score?

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

A. Lower

B. Higher

C. Closest to 100

D. Fewest runs

Answer: _____

2. (2 points) What is the Can Bonus for an Inside Can Distance of 40.0 cm?

A. -55 points

B. -70 points

C. -35 points

D. -20 points

Answer: _____

3. (2 points) Which brake power source is allowed?

Unit: Automatic stopping.
Worked-case data: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?
Separate investigation observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

A. A remote control

B. A tether from the start line

C. A pre-loaded rubber band that only operates the brake

D. A battery-powered motor

Answer: _____

4. (2 points) Why might a team use less than 2.00 kg?

A. It increases distance

B. Heavier masses are slower

C. Less energy makes stopping gentler and protects the egg

D. The rules require less

Answer: _____

5. (2 points) The vehicle does not move when triggered. What happens?

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

A. A run has not occurred; the team may keep working

B. It is a Failed Run

C. It scores 2500

D. The team is disqualified

Answer: _____

6. (4 points) Why is the egg required to be foremost?

Unit: Protecting the egg.
Worked-case data: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.
Separate investigation observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

Answer: _____
_____
_____

7. (4 points) How can a team check alignment quickly?

Unit: Drivetrains: turning a drop into distance.
Worked-case data: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?
Separate investigation observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

Answer: _____
_____
_____

8. (4 points) Why must the falling mass not hit the floor?

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

Answer: _____
_____
_____

9. (4 points) How can a vehicle be made to follow a gentle arc?

Unit: The Can Bonus and curved paths.
Worked-case data: Target 9.00 m. The team sets a 60 cm gap. Find the bonus, the gap center offset, and the arc radius.
Separate investigation observations:
Arc runs: Passed between cans 10 of 10.
Stopping: Stopped 18 cm short of the target each time.
Setting: Turns calculated from the straight-line distance.

Answer: _____
_____
_____

10. (4 points) Why measure Vehicle Distance from the egg tip?

Unit: Automatic stopping.
Worked-case data: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?
Separate investigation observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A 1.50 kg mass falls 0.60 m. If 40 percent reaches a 0.50 kg vehicle as kinetic energy, what is its top speed?

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.

Answer: _____
_____
_____

13. (2 points) Why did Design B protect the eggs?

Supplied synthetic observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

A. Its deceleration was about 10 times lower

B. It was heavier

C. It traveled farther

Answer: _____

14. (2 points) Define “Drive axle” in the context of Scrambler.

Answer: _____
_____
_____

15. (2 points) Define “Terminal Barrier” in the context of Scrambler.

Answer: _____
_____
_____

16. (2 points) Define “Outside Can” in the context of Scrambler.

Answer: _____
_____
_____

17. (2 points) Define “Wing nut” in the context of Scrambler.

Answer: _____
_____
_____

18. (2 points) Define “Rolling resistance” in the context of Scrambler.

Answer: _____
_____
_____